Venkata Lakshmi Parimala Pasupuleti

Forward Deployed AI Engineer · Agent Architect · Enterprise AI Transformation · LangGraph · MCP · Workflow Automation · Python · FastAPI · Prototype to Production

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PROFESSIONAL SUMMARY

Software engineer and AI builder with 8 years shipping production systems (100K+ MAU, 1M+ records) and hands-on experience turning ambiguous business problems into reliable agent workflows. Built **Leasel** - started as a vague support bottleneck (eligibility investigations taking a full day with no repeatable process), ended as an **8-agent LangGraph StateGraph** running in production: durable state, conditional routing, HITL escalation gate, citation-grounded RAG, and a 5-metric eval harness that regression-tests against 1,240 cases in CI. That is exactly the prototype-to-production, ambiguity-to-architecture journey Arcesium describes. Built **AgentProbe** - a reusable, zero-cost AI evaluation toolkit with templates, a CI regression harness, and playbooks that any team can use to ship LLM-powered workflows safely. At Entrata (8yr): shipped 20+ internal tools and workflow automations for operations teams, eliminated 80% of manual workload via Python/Bash pipelines, and maintained financial compliance systems where one wrong record affected thousands of residents' eligibility. Strong on APIs, systems design, and data - the engineering fundamentals that make agent workflows reliable in production. Case study: github.com/venkatalakshmiparimala/AgentProbe

CORE COMPETENCIES

Forward Deployed Engineering | Enterprise AI Transformation | Business Workflow Automation | Agent Frameworks / LangGraph | MCP Servers | LLMs / Tool-Calling / Structured Outputs | Agentic Workflows | Prototype to Production | Build-vs-Buy Analysis | Reusable AI Components & Playbooks | KPI Definition & Impact Measurement | Systems Design / APIs / Data | Internal Tooling for Business Teams | RAG Pipelines / Hybrid Search | HITL / Human-in-the-Loop | Iterative Delivery | Python / FastAPI | AWS / Azure | PostgreSQL / Snowflake / SQLite | CI/CD (GitHub Actions) | Docker / Kubernetes | Multi-Tenant Architecture | Scalable & Maintainable Solutions | Ambiguity & Ownership | RabbitMQ / Celery / Redis

EXPERIENCE

Blue Horse Digital

Sep 2026 - Present

AI/ML Engineer (Volunteer) — Virginia, USA

- Building ML models and data analysis pipelines; integrating AI components with backend/frontend teams; surfacing insights from raw business data into application-facing APIs and reporting layers.

Magizhs Tech Solution

May 2025 - Aug 2025

AI/ML Engineer Intern

- Built an **agentic mission-planning system** as a LangGraph stateful graph (parse → plan → retrieve → validate → conditional routing) - delivering a risk-aware compliance strategy: compliant missions export to QGroundControl; non-compliant ones blocked with cited FAA regulation.
- Built and deployed an **MCP server** for operator tool integration, enabling downstream tools to query the planning system via Model Context Protocol; implemented a RAG layer over FAA Part 107 corpus with guardrails enforcing altitude limits, battery budgets, and no-fly-zone checks.

George Mason University

Aug 2024 - Aug 2025

Research Associate / Graduate Teaching Assistant — Virginia, USA

- Built multi-source prediction models (Python, R) on WHO and UK Road Safety datasets across 190+ countries - R^2=0.977; applied Isolation Forest anomaly detection across 3 research pipelines, surfacing anomalies manual review missed.
- Fine-tuned a custom classification model using HuggingFace Transformers + PyTorch; deployed inference API on Quart; presented model design and evaluation results to faculty leadership.
- Delivered technical workshops to 600+ students (Python programming, database design, SQL) across multiple MIS 303 sections.

Entrata - PropTech / Real Estate SaaS

Sep 2018 - Aug 2024

Senior Software Engineer — Lehi, UT (Remote)

- **Shipped 20+ internal tools and workflow automations** used by operations, compliance, and support teams across 120+ Agile sprints - reduced manual operational workload by 80% via Python/Bash automation pipelines; tools ran in production for 100K+ monthly active users across 1M+ property records.
- **Owned the HUD income-limit pipeline for 7 years** - a few thousand federal rows expanded into ~180,000 domain records powering eligibility, rent, and HAP calculations; designed baseline validation and rollback readiness for every release; zero compliance violations across ownership period.
- **Built RabbitMQ-based async workflow automation** for gross-rent updates across thousands of tenant records - designed retry logic and idempotency guarantees eliminating double-processing failures; owned through production incidents and annual schedule changes.
- **Led AWS S3 data migration with 100% PHI compliance** - Libsodium-encrypted pipelines, zero security incidents; optimized 200+ SQL queries (30% performance gain) across complex multi-tenant financial schemas.
- Led microservices redesign using Docker - 50% scalability improvement, 30% deployment time reduction; mentored 5+ engineers; conducted 300+ peer code reviews; built internal debugging tools cutting debug time by 40%.

Entrata India - Db Xento Systems

Jan 2017 - Sep 2018

Associate / Trainee Software Engineer — Pune, India

- Built property management and compliance modules (PHP, Python, Java, PostgreSQL); led AWS S3 cloud migration of client documents; designed and migrated database schema for hundreds of thousands of records.

PROJECTS

Leasel: Agentic AI Support Investigation Platform

LangGraph · 8-Agent · HITL · RAG · Prototype to Production

Started with an ambiguous business problem: support engineers spending a full day investigating eligibility tickets because the root cause was always scattered across settings, regulations, historical incidents, and code. Converted that into a structured problem statement, designed the agent architecture, and shipped a production platform. **8-agent LangGraph StateGraph** with durable state (Triage → Retrieval → Configuration → Historical Incident → Structured Data → Reasoning → Validation → Human Escalation) - **conditional HITL gate** fires when confidence is too low; LLM extracts and reasons, deterministic logic makes the final regulated call; hybrid RAG (TF-IDF + semantic reranking) with citation grounding to exact source clauses. Defined KPIs and measured impact: 93.6% precision, 87.9% F1 on 1,240 test cases; pass@1 81.7% via CI regression harness.

Python · LangGraph (8-agent StateGraph) · FastAPI · TF-IDF · Semantic Reranking · OCR · SQLite · 2026

AgentProbe: Reusable AI Evaluation Toolkit

Reusable Components · Templates · Playbooks · CI/CD

Built to solve the reusability problem: every LLM project rebuilding evaluation from scratch with no shared methodology. **Reusable templates, regression harnesses, and playbooks** any team can use to ship LLM-powered workflows safely - 15-endpoint FastAPI backend, **5-metric eval engine** (correctness, groundedness, latency, token efficiency, hallucination rate), CI-integrated golden-dataset harness, model routing across OpenAI/Anthropic/Groq/Ollama, offline + online evaluation loops. **Build-vs-buy decision made explicit:** built from scratch to eliminate vendor dependency and enable zero-cost local execution. Used to evaluate Leasel across 1,240+ test cases. Live: github.com/venkatalakshmiparimala/AgentProbe

Python · FastAPI · LangGraph · ChromaDB · CI/CD (GitHub Actions) · OpenAI / Anthropic / Groq / Ollama · 2025

Agentic Drone Mission Planner + MCP Server (Magizhs)

MCP · LangGraph · Agent Frameworks · FAA Compliance

Translated natural-language survey requests from marketing, real estate, and commercial stakeholders into structured flight plans using a LangGraph stateful graph (parse → plan → retrieve → validate → export). Built an **MCP server** enabling downstream operator tools to query the planning system via Model Context Protocol. RAG layer over FAA Part 107 corpus with deterministic guardrails (altitude limits, battery budgets, no-fly-zone geometry) blocking non-compliant missions before MAVLink/QGroundControl export.

Python · LangGraph · FastAPI · MCP · RAG (FAA Part 107) · Photogrammetry math · 2025

EDUCATION

George Mason University **- M.S. Data Analytics Engineering** May 2026

GPA: 3.87 · Virginia, USA · Coursework: ML systems, NLP, cloud infrastructure, statistical modeling, deep learning

Jawaharlal Nehru Technological University **- B.Tech Computer Science & Engineering** 2017